

Final Project Review Presentation Script

& Electrical Engineering Defense Dossier

ENERGY DEMAND FORECASTING OF SMART GRID USING DEEP LEARNING

Course: 23EER705 Research Based Mini Project / Capstone	Project Group: ER27A15 • First Review
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1. Presentation Roadmap & Timing Schedule

The First Review presentation strictly permits 10 to 12 minutes for presentation delivery, followed by 5 minutes of intensive technical viva by the EEE/ER evaluation panel. Slides 14 through 20 form the engineering core and forward roadmap of your defense.

Slide #	Slide Title & Header	Key Engineering Focus	Target Time
Slide 14	Simulation / Development Results (13 / 20)	8.8× Parquet compression, 99.9% outlier cap, scaling	60–75 sec
Slide 15	Results, Analysis & Validation (14 / 20)	Data quality, feature readiness, Phase-I status	60–75 sec
Slide 16	OBE, SDG & Traceability (15 / 20)	Mapping to UN SDG 7, CO1, CO4 & SPI-TP-01	45–50 sec
Slide 17	Publication & Deliverable Plan (16 / 20)	IEEE Conference paper, software copyright	40–45 sec
Slide 18	Challenges, Risks & Actions (17 / 20)	Data variance, computational load, guide input	45–50 sec
Slide 19	Plan Before Next Review (18 / 20)	Execution of WP3, WP4, WP5 (Model training)	45–50 sec
Slide 20	Conclusion / Review Summary (19 / 20)	60-second executive recap & immediate next steps	50–60 sec
Slide 21–22	References & Thank You / Q&A	Literature citations & opening for panel inquiry	Transition

2. Word-for-Word Spoken Presentation Script

Slide 14: Simulation / Modelling / Development Results (13 / 20)

Timing: 70 sec

Evidence: E2, E3 (WP2/WP3)

Visual Cue: Two-panel figure on left (storage compression + signal normalization), Result Details table on right.

• Presenter Speaking Script

“Respected evaluators, moving on to Slide 14, we present the strongest actual technical and engineering result achieved during this First Review phase, traced under Work Packages 2 and 3 with Evidence Codes E2 and E3.

[Point to Left Chart — Panel A] One of the primary engineering bottlenecks in smart grid analytics is handling massive high-frequency data streams. Our raw smart meter dataset contains over 322 million interval readings, expanding to over 37 gigabytes uncompressed. To enable high-throughput training without memory crashes, we engineered a chunked binary ingestion pipeline using Apache Arrow Parquet with Snappy compression. As illustrated in the left bar chart, this achieved an 8.8-times storage reduction—compressing 37.2 gigabytes down to just 4.1 gigabytes—while accelerating disk input/output read throughput by more than 10 times.

[Point to Left Chart — Panel B] On the signal processing front, shown in Panel B, raw smart meter series suffer from severe sensor noise, meter transmission errors, and negative values. We implemented an automated zero-loss cleaning pipeline: negative meter errors are clipped to zero, and extreme sensor spikes are bounded at the 99.9th percentile threshold (2.929 kilowatt-hours) to prevent gradient explosion during backpropagation. Finally, min-max scaling normalizes the load demand into a continuous [0, 1] range, ensuring optimal convergence for deep neural networks.

As summarized in the table on the right, all 322 million records have achieved 100% data continuity, over 20 temporal and cyclical features have been engineered, and our data engineering foundation is fully verified.”

Slide 15: Results, Analysis & Preliminary Validation (14 / 20)

Timing: 65 sec

Visual Cue: Four-row benchmarking table tracking Data Quality, Feature Readiness, Accuracy, and Generalization.

• Presenter Speaking Script

“On Slide 15, we present the preliminary validation and gap analysis against our approved project targets.

First, for **Data Quality**: Against a noisy baseline of raw CSVs with missing timestamps and duplicate records, our target was to produce a clean, continuous time-series tensor. As shown, our data preprocessing is 100% complete, meaning the dataset is fully structured and prepared for neural modeling.

Second, for **Feature Readiness**: Moving beyond raw scalar consumption values, we have extracted multidimensional features including continuous sine and cosine cyclical encodings for diurnal and seasonal periodicity, day-of-week indicators, and historical lag windows.

For **Forecasting Accuracy and Model Generalization**: As noted in our First Review schedule, these are marked as ‘Not yet evaluated’ because model-level training on the hybrid architecture is scheduled for Phase II under Work Package 4. Our approved benchmark is to achieve a superior Mean Absolute Percentage Error (MAPE) under 4% compared to conventional baseline models.

To summarize the bottom panel: What is **Verified Now** is the complete dataset ingestion, cleaning, and feature tensor preparation. What remains **Still to Validate** in the next phase is the deep learning model training,

hyperparameter convergence, and cross-dataset evaluation.”

Slide 16: OBE, SDG & Sustainability Traceability (15 / 20)

Timing: 45 sec

Mapping: UN SDG 7 • CO1, CO4 • SPI-TP-01

Visual Cue: Traceability matrix linking Objectives, COs, SDGs, and SPI Codes.

• Presenter Speaking Script

“Slide 16 establishes our compliance with Outcome-Based Education (OBE) and United Nations Sustainable Development Goals.

Our project directly maps to **UN SDG 7: Affordable and Clean Energy**. In modern smart grids, high forecasting errors force utilities to run inefficient, fossil-fuel-based spinning reserves to handle unexpected demand surges. By developing a high-fidelity deep learning forecasting model, our solution optimizes spinning reserve margins, lowers operating costs, and facilitates the reliable integration of variable renewable energy sources.

From an academic standpoint, Objective 1 and Objective 2 directly address **Course Outcome CO1**—applying core electrical engineering and computational principles to prepare data and formulate architectures. Objectives 3 and 4 map to **CO4**—conducting formal investigations of complex engineering problems through comparative evaluation using standardized metrics like MAE, RMSE, and MAPE under performance indicator SPI-TP-01.

As shown in the status column, Evidence E1 and E5 are completed, while E7 and E8 will document our model training and comparative results in Phase II.”

Slide 17: Publication, IPR & Technology Deliverable Plan (16 / 20)

Timing: 40 sec

Visual Cue: Table detailing deliverables; primary deliverable checkbox ticked at Conference Paper.

• Presenter Speaking Script

“Turning to Slide 17, we outline our publication and technology deliverable plan.

Our **Primary Target Deliverable** is a peer-reviewed **Conference Paper** submitted to an IEEE or Scopus-indexed conference, such as IEEE PES or an international electrical energy conference.

The specific publishable contribution of this work will be the novel architectural formulation of our hybrid deep learning model and its validated performance against baseline predictors across diverse smart grid load profiles.

Our immediate action item is to complete the model training and evaluation pipeline to generate publishable comparative figures. An extended journal manuscript and software copyright documentation for our automated preprocessing codebase will be assessed based on the final experimental results.”

Slide 18: Challenges, Risks & Corrective Actions (17 / 20)

Timing: 50 sec

Visual Cue: Four technical risks and proactive engineering mitigations.

• Presenter Speaking Script

“On Slide 18, we highlight the major technical risks identified and the proactive engineering mitigations our team has established:

1. Dataset Variability: Real-world load profiles differ drastically across residential smart meters and bulk regional grids. To prevent model divergence, we have implemented standardized normalization and uniform sliding-window batch tensors.

2. Hybrid Model Architecture Complexity: Selecting an optimal combination of recurrent and attention mechanisms can introduce development delays. We mitigated this by benchmarking established architectures from recent IEEE literature.

3. High Computational Demand: Training recurrent networks across hundreds of millions of sequence steps requires significant memory. We have resolved this by implementing chunked data generators and columnar Parquet streaming to minimize RAM pressure.

4. Model Generalization Across Datasets: To prevent overfitting to a single utility, our validation protocol tests the model across both the Australian SGSC dataset and American PJM regional datasets.

We have designated technical input from our project guides as our primary support requirement to finalize hyperparameter bounds.”

Slide 19: Plan Before Next Review (18 / 20)

Timing: 45 sec

Visual Cue: Execution roadmap for WP3, WP4, WP5.

• Presenter Speaking Script

“Slide 19 outlines our execution roadmap leading into the Second Project Review. Our next major milestone is to have a **fully trained and evaluated hybrid deep learning forecasting model**.

Specifically, our team will execute three sequential work packages:

- Under **Work Package 3**, we will complete the PyTorch implementation of the hybrid architecture (Evidence E5).
- Under **Work Package 4**, we will conduct end-to-end model training, hyperparameter tuning, and calculate our core performance metrics—MAE, RMSE, and MAPE—on unseen test sets (Evidence E5 and E7).
- Under **Work Package 5**, we will conduct baseline comparison against standard benchmarks (such as ARIMA, standalone LSTM, and GRU) to quantify our engineering improvements (Evidence E8).

Concurrently, we will document these experimental findings to prepare our draft research paper.”

Slide 20: Conclusion / Review Summary (19 / 20)

Timing: 50 sec

Visual Cue: 6-part executive summary card.

• Presenter Speaking Script

“To conclude our First Project Review on Slide 20:

- The **Engineering Problem** we are addressing is the high non-linearity and volatility of smart grid energy demand, which conventional forecasting methods fail to accurately track during critical peak windows.

- Our **Proposed Engineering Contribution** is to build a robust hybrid deep-learning forecasting framework capable of learning temporal dependencies and non-linear demand patterns.
- During this phase, our **Major Work Completed** includes a comprehensive literature survey, selection of benchmark smart grid datasets, an automated data cleaning pipeline, and multi-scale feature engineering.
- Our **Most Significant Result** is the complete validation and 8.8-times storage compression of the SGSC dataset, preparing clean, continuous tensors for deep learning execution.
- Our **Immediate Next Target** is to commence training of the hybrid deep learning model and validate its forecast accuracy against approved benchmarks.”

Slide 21 & 22: References & Concluding Remarks

• Presenter Speaking Script

“[Advance to Slide 21] Our methodology is grounded in recent IEEE Transactions on Smart Grid, IEEE Access, and IEEE Open Journal literature published between 2022 and 2026.

[Advance to Slide 22] Thank you, respected members of the evaluation panel, for your time and attention. We now welcome your valuable questions, feedback, and guidance.”

3. Electrical Engineering Defense Dossier (Viva Q&A)

The following questions represent core power systems and electrical grid concepts that EEE faculty evaluators routinely ask to verify physical domain understanding.

Q1: Why do electrical power utilities need Short-Term Load Forecasting (STLF)?

Answer: Primarily for **Unit Commitment** and **Economic Load Dispatch**. Thermal generating stations require 4 to 8 hours to ramp up from cold start. Day-ahead load forecasting allows the system operator to commit the optimal combination of generating units to meet expected demand at minimum operating cost while satisfying spinning reserve requirements.

Q2: What is the physical consequence of Under-Forecasting vs. Over-Forecasting on the power grid?

Answer:

- **Under-Forecasting (Threat to Stability):** Generated power is less than demand ($P_G < P_D$). The deficiency draws kinetic energy from generator rotors, causing system frequency to drop below 50 Hz. This risks tripping under-frequency relays, causing localized load shedding or cascading blackouts. Emergency peaker plants must be dispatched at high expense.
- **Over-Forecasting (Economic Penalty):** Generated power exceeds demand ($P_G > P_D$). System frequency rises, forcing thermal generators to operate throttled back at poor thermal heat rates, leading to fuel wastage and high operating expenses.

Q3: What forecasting horizon does this project target?

Answer: **Short-Term Load Forecasting (STLF)**, spanning 30-minute intervals up to 24–48 hours ahead (day-ahead scheduling). In contrast, **Very Short-Term** (seconds to minutes) is used for AGC/frequency control; **Medium-Term** (weeks) for maintenance scheduling; and **Long-Term** (years) for transmission expansion planning.

Q4: Are you forecasting Active Power (kW/MW) or Energy (kWh)?

Answer: In the **SGSC smart meter dataset**, we forecast energy consumption per 30-minute interval (kWh), which is proportional to average active power over the half-hour interval ($P_{\text{avg}} = \text{kWh} / 0.5 \text{ h} = 2 \times \text{kWh in kW}$). In the **PJM dataset**, we forecast hourly aggregate active power demand directly in Megawatts (MW).

Q5: Why is household smart-meter load significantly more volatile than substation load?

Answer: Due to the **diversity factor** and **aggregation effect**. At the regional transmission level (PJM), millions of individual consumer demands aggregate together; non-coincident peak demands cancel out stochastic variations according to the Central Limit Theorem. At the single household level (SGSC), individual appliance switching (e.g., 3.5 kW HVAC inrush, water heaters, EV charging) causes sudden step-changes in load with high variance.

Q6: What is the “Duck Curve”, and how does your model handle it?

Answer: The Duck Curve represents the deep midday trough in net load caused by high distributed solar PV generation, followed by a steep evening ramp when solar generation drops just as residential consumption surges. Our feature engineering captures this by transforming timestamps into continuous cyclical sin and cos encodings, enabling the model to learn diurnal solar-load ramp transitions.

Q7: What physical systems do the SGSC and PJM datasets represent?

Answer:

- **SGSC (Smart Grid Smart City, Australia):** Low-voltage (230 V) residential distribution network smart meter readings at 30-minute intervals. Reflects consumer-level demand dynamics.
- **PJM Interconnection (USA):** Bulk high-voltage transmission grid (138 kV to 765 kV) across 13 eastern US states. Reflects regional aggregated grid demand.

Q8: Why are MAPE and RMSE preferred over MSE in power systems?

Answer:

- **MAPE (%):** Being dimensionless, it allows grid engineers to benchmark performance across different feeders and regional substations of varying base load sizes.
- **RMSE (kW/MW):** Penalizes large error outliers quadratically. In power systems, a single large forecast error during peak load creates severe reserve shortfalls, whereas small steady-state errors are easily managed by governor action.

Q9: Why use Min-Max normalization instead of Standard (Z-score) scaling?

Answer: Electrical load has a strict physical lower boundary at zero ($\geq 0 \text{ kW}$). Standard scaling produces negative normalized values that have no physical meaning for consumption and can disrupt neural activations. Min-Max normalization bounds inputs within $[0, 1]$, matching sigmoid/tanh gate characteristics and preventing exploding gradients.

Q10: 30-Second Golden Summary Statement (Core Electrical Engineering Value)

Core Statement: “Sir/Madam, from an electrical engineering perspective, this project is not merely an algorithmic exercise. We are developing an automated Short-Term Load Forecasting tool that enables smart grid operators to optimize Day-Ahead Unit Commitment and minimize spinning reserve margins. By reducing peak forecasting error on volatile distribution networks, our model prevents frequency instability, reduces expensive fossil peaker generation,

and directly supports the seamless integration of renewable energy under SDG 7.”

